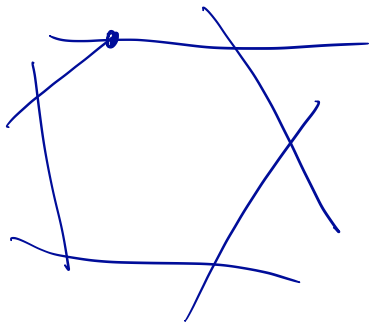


Linear Optimization and Extreme Points

$$\begin{array}{ll}\min: & c^T x \\ \text{st:} & Ax \leq b\end{array}$$

convex & opt. solns at
ext points ("corners")

which can encode combinatorial
structure.



Polytopes and Polyhedra

$$\underline{\mathbb{R}^n}$$

$$H = \{x \in \mathbb{R}^n : a^T x = b\} \quad \text{Hyperplane}$$

$$H_+ = \{x \in \mathbb{R}^n : a^T x \geq b\} \quad \text{Half-space}$$

$$P = \{x \in \mathbb{R}^n : Ax \geq b\} \quad A \text{ } m \times n, \quad b \in \mathbb{R}^m$$

All convex.

$$x_1, x_2 \in P$$

$$x_3 = \lambda x_1 + (1-\lambda)x_2 \quad \lambda \in [0,1]$$

$$Ax_3 = \lambda Ax_1 + (1-\lambda)Ax_2 \geq b.$$

Extreme Points



Thm: T.F.A.E.

1. $x \in \text{Ext } P$ if $\nexists y \neq z, y, z \in P$
 $x = \lambda y + (1-\lambda)z, \lambda \in (0,1)$

2. $x \in \text{Ext } P$ if $\exists c \in \mathbb{R}^n$ s.t. $c^T x < c^T y \forall y \in P, y \neq x$.

3. $x \in \text{Ext } P$ if n of the constraints are

independent and tight/binding/active
 $\{a_i\}$ independent
 $a_i^T x = b_i$
Basic Feasible Solution (BFS).

Standard Form Polyhedra

$$\text{"Ax} \leq b\text{"}$$

// Note: If P is
bounded it is a polytype.

$$\begin{cases} Ax = b \\ x \geq 0 \end{cases}$$

$$Ax \leq b$$



$$Ax + Is = b$$

$$s \geq 0$$

$$x_i = x_i^+ - x_i^-, \quad x_i^+, x_i^- \geq 0$$

$$Ax^+ - Ax^- + Is = b$$

$$x^+, x^-, s \geq 0$$

WLOG

$$Ax = b$$

$$x \geq 0$$

Assume rows of
 A are
independent.

Extreme Points and Standard Form

$$\underline{P} = \left\{ \begin{array}{l} Ax = b \\ x \geq 0 \end{array} \right\} \subseteq \mathbb{R}^n$$

$$A \quad m \times n$$

$$m \leq n$$

$x \in \text{Ext } P$ if we have n
tight - ind. constraints.

m come from $Ax = b$.

$n-m$ more? $x \geq 0$: $n-m$ of these must be
satisfied with equality.

$B \subseteq [n]$ indices of x that are possibly non-zero.

$$x = \begin{pmatrix} x_B \\ 0 \end{pmatrix}$$

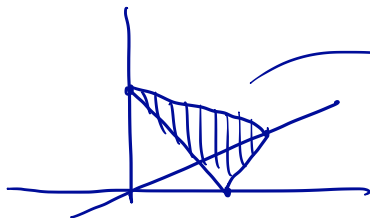
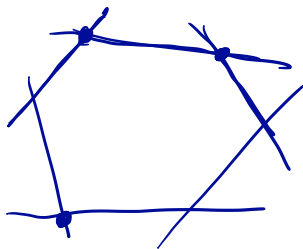
$$Ax = b$$

$$\begin{bmatrix} A_B & A_{B^c} \end{bmatrix} \begin{pmatrix} x_B \\ 0 \end{pmatrix} = b$$

$$\boxed{x_B = A_B^{-1} b}$$

Adjacent Extreme Points

By picture



$$\begin{aligned} a^T x &= 1 \\ x &\geq 0 \end{aligned}$$

in $\mathbb{R}^3$.

Adjacent Extreme Points and Bases

$$Ax = b$$
$$x \geq 0$$

$$x \in \text{Ext } P :$$

$$x = \begin{pmatrix} x_B \\ 0 \end{pmatrix},$$

$$B \subseteq \{1\}$$

$$|B| = m$$

↑ "Basis"

B_1, B_2 are "adjacent" if they share $m-1$ indices.

↓

$$x_1 = \begin{pmatrix} x_{B_1} \\ 0 \end{pmatrix}$$

$$x_2 = \begin{pmatrix} x_{B_2} \\ 0 \end{pmatrix}$$

$$|B_1 \cap B_2| = m-1$$

Q: How many neighbors can an extreme point have?

Convex Hull of Extreme Points

Given $x_1, \dots, x_d$

$$\begin{aligned}\text{Convex Hull} &= \text{Conv} \{x_1, \dots, x_d\} \\ &= \left\{ x = \sum_{i=1}^d \lambda_i x_i, \lambda_i \geq 0, \sum \lambda_i = 1 \right\}.\end{aligned}$$

Thm: If $P = \{x : Ax \leq b\}$ is bdd,

then $P = \text{Conv}(\text{Ext } P)$.

Optimality and Extreme Points

$$\min: c^T x$$

$$\text{st: } x \in \mathbb{P}, \quad \mathbb{P} \text{ bdd.}$$

Thm: If feasible, then $\exists$ an opt
sol'n at an ext. point.

Follows from prev. slide.

If $y \in \mathbb{P}$ is opt.

$$y \in \mathbb{P} \Rightarrow y = \sum \lambda_i z_i, \quad z_i \in \text{Ext } \mathbb{P}$$
$$c^T y = \sum \lambda_i c^T z_i$$